

pole-ai — Business Model Options

Monetization strategies for the Athlete Trick Identification System.

Compiled from docs/business/ · 14 documents.

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Monetization options for the already-built Athlete Trick Identification System. Every option below reuses **existing** packages/apps (`pole_ml` , `pole_tools` , `pole_chatbot` , `pole_api` , `pole_fe` , `pole_analyst` , `jobs` , `pole_crop` , Keycloak identity, k3s/Helm infra). Each strategy is documented per type under this folder with: how it makes money, subscription vs per-service pricing, and the implementation surface (new app, API-only, frontend-only, ...).

Comparison Matrix

#	Type	Option	Monetization model	Pricing	New app?	Effort	Priority
S1	subscription	Personal trainer (tiers)	Subscription (B2C)	€9-49 /mo tiers	pole_fe upgrades	Medium	★★★★
S2	subscription	Coach / gym SaaS	Subscription (B2B seats)	€49-199 /mo per studio	pole_fe + multi-tenant	Medium	★★★★
S3	subscription	Deep-report upgrade (freemium)	Freemium subscription	Free + €7 /mo	pole_fe (paywall)	Low	★★
A1	api	Chatbot pay-per-query	Metered API (B2B)	€0.001-0.01 /query	API-keys + billing	Low	★★★★
A2	api	B2B video-analysis API	Metered API (B2B)	€0.05-0.20 /video	API-keys + billing	Low	★★★★
P1	per-service	Pay-per-video credits	Credits (B2C)	€2-5 /analysis, packs	pole_fe (checkout)	Low	★★
P2	per-service	Grade / level assessment	One-off (B2C)	€19-39 /assessment	classifier + curriculum DB	Medium	★★
L1	licensing	White-label studio license	License fee (B2B)	€1-5k /yr	infra multi-tenant	High	★★
L2	licensing	Named-trick dataset library	Dataset license (B2B)	€500-5k one-off	export + data contract	Medium	★
M1	marketplace	Coach lead-generation	Lead fee / listing (B2B)	€20-50 /lead	listing app	High	★
M2	marketplace	Verified athlete profiles	Freemium badges (B2C)	€4 /mo badges	profile + verify flow	Medium	★
E1	events	Virtual trick challenges	Entry fee + sponsors (B2C)	€5-15 /event	event engine	Medium	★★

How to read each option

Every strategy file follows the same template:

1. **What it is** — one-line product description.
2. **How it makes money** — the revenue mechanism + pricing rationale.
3. **Subscription or per-service** — explicit, with a tier/price suggestion.
4. **Implementation** — surface (new app vs API-only vs frontend-only), what is **already built** (reuse), and what must be **added** (gaps).
5. **Effort & priority** — relative build cost and go-to-market sequencing.

Quick guidance

- **Fastest to ship (hours, not weeks): A1 / A2 / P1** — they reuse the existing `pole_api` + `pole_chatbot` + jobs infra with an API-key + metering layer.
- **Best moat / defensibility: P2, S2, S1** — they sit on the trained LSTM + cohort histogram engine that competitors would have to re-train from data.
- **Lowest marginal cost to operate: A1/A2** — per-query billing on already-costed inference (LLM tokens + media processing are the variable costs to meter).
- **Realistic first launch: ship A1 (chatbot API) + P1 (per-video credits)** with the same billing rail, then layer **S1/S2 subscriptions** behind Keycloak.

See also: `docs/packages/chatbot/PLAN.md`, `docs/app/pole_api/PLAN.md`,
`docs/app/pole_fe/PLAN.md`, `docs/app/keycloak/PLAN.md`, `docs/packages/pole_ml/PLAN.md`.

[Market Research] `pole-ai` business models

What this file is. A fast, first-pass market sizing for the 12 monetization options in this folder. For each idea it answers two business questions: who is the target, and roughly how many people could realistically become customers. Estimated from public 2025–26 industry data; ranges are deliberately wide and each option ends with the check that tightens the estimate.

How market research works

You don't need a data team. You need a **funnel** with three numbers and two checks.

The funnel: TAM → SAM → SOM

Term	Meaning	In one line
TAM	Total Addressable Market	Everyone who could ever use it
SAM	Serviceable Addressable Market	The slice you can actually reach (language, region, platform)
SOM	Serviceable Obtainable Market	Customers you can realistically win in ~3 years — the number that matters

As a small business, assume you capture **0.5–3% of SAM**. Optimistic marketing can stretch that; a B2B niche with few competitors can beat it on revenue per customer.

The two checks

1. **Willingness to pay** — look at what 5–10 real competitors charge. Their prices *are* the market's answer; you price around them.
2. **Bottom-up validation** — shrink the problem: talk to 10 coaches, run a waitlist landing page, count every studio in one region with a directory. Regret-direction: an estimate an order of magnitude wrong only costs you a week to correct when you start small.

What follows is the **top-down** half. Treat it as planning input, then run the bottom-up checks before building anything.

Anchor numbers (data used)

Fact	Value	Source
Pole-fitness equipment market	\$520M (2024) → \$1.13B (2033), ~9.1% CAGR	Research Intello, 2025
US audience actively searching pole-dance	~2.1M people (≈20% of global → ~10M "interested")	Rascasse audience data, 2026
Active practitioners worldwide (training, not just curious)	est. 1–3M (derived from studios + equipment + audience)	reasonable bounds
"Serious" athletes (train at home, track progress)	est. 10% of practitioners → 100–300k	assumption
Competitive athletes	3,000+ compete in PSO events yearly; IPSF worlds from 40+ countries; >200 athletes at German nationals	PSO, IPSF, polecamps
Studios + independent coaches	est. 2–5k studios + 5–10k freelancers (Mexico alone 400+; US ~1k+; "thousands" worldwide)	yourpolepal, PoleLife, IPIA
B2B benchmark pool (video-analysis APIs)	Hundreds of sport/fitness apps globally (Hudl/Dartfish-class demand)	industry

Why the ranges? "Interested" (~10M) is much bigger than "practices" (1–3M), which is bigger than "would pay" (tens of thousands to a few hundred thousand). Every row below keeps that gap explicit.

Per idea

Idea	Target public	Interested pool (SAM)	Realistic goal (SOM, yr 3)	Validate by
S1 — Personal trainer (subs)	Serious athletes training at home	50–150k reachable (of 100–300k serious)	1–3k subscribers (~€20–65k/mo)	Waitlist; compare workout-app churn
S2 — Coach / gym SaaS	Studios (2–5k) + freelancers (5–10k)	7–15k orgs	100–500 studios (~€15–75k/mo)	Interview 10 coaches; count studios locally
S3 — Deep-report freemium	Everyone doing trick ID (free) → upgraders	50–150k engaged free users	2–5k paying (2–5% conversion)	A/B a paywall on a landing page
A1 — Chatbot pay-per-query	App/platform developers needing video analysis	Hundreds of apps globally; pole niche <50	5–30 API customers	Survey fitness-app teams; check API signups
A2 — B2B video-analysis API	Sports/fitness apps + media	Same pool as A1	5–25 customers (~€5–50k/mo)	Publish API docs; watch signup rate
P1 — Per-video credits	Casual practitioners (the 90%)	900k–2.7M casual	1–3k paying/yr (~€10–50k)	Pre-sell "pay as you use"
P2 — Grade / level assessment	Syllabus- and competition-oriented athletes	20–50k (serious + competitive)	500–5k assessments/yr	Partner with a federation; price test
L1 — White-label license	Studios, chains, federations that can afford it (€1.5–5k)	500–1.5k orgs	5–30 licenses (~€8–150k/yr)	Pitch 3 studios before building
L2 — Dataset library	Sports-tech companies, researchers	Tens of buyers globally	1–10 deals	Verify data rights FIRST (legal)
M1 — Coach lead-generation	Coaches/studios near athletes	Your city/region only (hundreds)	10–50 paid accounts (needs density!)	Pick 1 metro; count studios there
M2 — Verified athlete profiles	Ambitious athletes (social proof)	100–300k serious	1–5k paid badges	Check engagement in pole FB/IG groups
E1 — Virtual trick challenges	Community + competitors + sponsors	20–50k engaged athletes	500–5k entries/yr + 2–20 sponsors	Run one pilot event free

What this tells you

1. **The niche is small but real and growing (~9% CAGR).** No idea here is a "hundreds of millions" play — but several are very healthy for a 1–3 person product.

2. **Highest revenue per customer:** S2, L1, P2 (B2B / services) — fewer customers, more money each.
 3. **Largest raw pools:** S1 / S3 / P1 (B2C volume) — with the *hardest* problem being marketing (reaching even 2% of a niche).
 4. **B2B API (A1/A2):** the pool is small but buyers churn the least once integrated — the classic "10 customers who pay forever" pattern.
 5. **Most realistic first move, by data: P2 (assessment)** if you can partner with a federation, or **A1 / P1** for the fastest self-serve revenue. SOM is small everywhere, so winning = distribution, not just the build.
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Sources

- Research Intello — *Pole Fitness Pole Market Report 2033 (2025)*.
- Rascasse — *Pole dance: 2.1M Fans in the United States (2026)*.
- Pole Sport Organization — *How It Works* (3,000+ competitors/year).
- IPSF / Wikipedia — *Pole sports* (federation, world championships).
- yourpolepal — *Worldwide Pole Studio Directory* (Mexico 400+ studios).
- PoleLife — *Pole studio directory* (289 US studios).
- IPIA — *Pole Industry Financial Survey (2024/2026)*.
- polecamps — *A guide to pole competitions worldwide (2026)*.

Estimates are the author's own; ranges were chosen to be directionally safe rather than precise. Last updated 2026-09-06.

[S1] Personal Trainer (Subscription Tiers)

Type: subscription/ · B2C recurring revenue · Priority ★★ ★

Description

The idea in plain English.

Trainers are expensive and don't watch you between classes. This product is an "AI coach in your pocket": an athlete films herself doing a trick (e.g. a shoulder mount), and the app recognizes the move and acts like a personal trainer — it tells her how her execution compares to other athletes, highlights frames where her form can improve, and lays out a simple weekly training plan.

How it makes money.

Monthly subscription with a free plan to attract users, then paid tiers:

- **Free** — you get your trick identified, once a week. Enough to taste the product.
- **Plus (€9/month)** — unlimited identification + detailed video feedback.
- **Pro (€19/month)** — full comparison against other athletes + monthly training plans.
- **Studio (€49/month)** — for coaches/gyms managing a whole group.

Recurring revenue: once an athlete is hooked on tracking her progress, she stays subscribed month after month.

Target public.

Pole fitness athletes — from beginners who just landed their first trick to semi-competitive athletes — and the coaches/studios who train them. People who already spend €30–80/month on classes and value any tool that makes their hidden practice time count.

What it is

A consumer-grade "AI personal trainer" for pole athletes: they record a trick, the app identifies it, and a monthly subscription unlocks automated coaching — cohort percentile scores, critical-frame critiques, and a 4-week improvement plan.

How it makes money

Recurring monthly subscription with a free tier as acquisition. The value is "an always-available coach between lessons" — the LLM tips + cohort analytics are the differentiating content that users pay to keep seeing.

Subscription or per-service?

Subscription. Pricing suggestion (EU market):

Tier	Price	What unlocks
Free	€0	Trick ID + 1 analysis / week, no reports
Plus	€9 /mo	Unlimited ID, deep-report, 10 critical-frame critiques
Pro	€19 /mo	Cohort percentile, 4-week plans, multi-athlete + history
Studio	€49 /mo	Everything + team/class profile (see S2)

VAT/region pricing and an annual (-20%) option are the normal add-ons.

Implementation

Surface: frontend-first (`pole_fe` / `pole_analyst` upgrades) + billing layer. No new ML.

Already built (reuse):

- `/packages/pole_ml` — LSTM classifier + embeddings + hybrid Chroma fallback.
- Histogram/cohort z-score engine and critical-frame extraction (8-signal analysis).
- `pole_chatbot` + `pole_api` — coaching feedback flow (z-score outliers → LLM tip / 4-week plan).
- Keycloak identity — magic links, temp-access sessions, GDPR purge (consumer-ready auth).
- `pole_fe` / `pole_analyst` — Angular frontends, design-system tokens, model registry.

Must add (gaps):

- Billing integration (Stripe / Paddle) + webhooks.
- Tier gating on the API layer (`pole_api` slice: `billing`).
- Paywalled report generation with usage counters (meter per analysis).
- Free-tier weekly usage cap (reuse the sliding-window rate limiter already in `pole_chatbot`).

Effort & priority

Effort: Medium. No model work; the work is checkout, gating, and report UX.

Priority: ★★★★★ — it monetizes the coaching value already built and is the natural B2C flagship.

See also: `docs/app/pole_analyst/PLAN.md` , `docs/packages/pole_ml/PLAN.md` ,
`docs/app/pole_fe/PLAN.md` , `docs/app/keycloak/PLAN.md` .

[S2] Coach / Gym SaaS (Seat License)

Type: subscription/ · B2B recurring revenue · Priority ★★ ★

Description

The idea in plain English.

A dashboard for pole coaches and gyms to manage their whole team in one place: each athlete's tricks, how they compare with each other, and automatic alerts when someone is improving or stalling — even for athletes training at home. The coach suddenly sees the entire roster without filming anything himself.

How it makes money.

Per-studio monthly subscription:

- **Coach Solo (€49/month)** — one coach, up to 20 athletes.
- **Gym (€149/month)** — 3 coaches, 100 athletes, team comparisons.
- **Gym+ (€299/month)** — unlimited, white-label reports, priority support.

Recurring B2B revenue: once a studio builds years of athlete history on the platform, it is very hard to leave.

Target public.

Pole studios, gyms, and independent coaches who train multiple athletes and already pay for booking/video tools. They buy the "full roster visibility" that a human coach can't get alone.

What it is

A per-studio subscription for pole coaches and gyms: the coach manages a **roster of athletes**, tracks their tricks, and gets cross-athlete cohort analytics and progress alerts. Athletes film at the studio or at home; the coach sees everything in one dashboard.

How it makes money

B2B monthly seat/studio pricing. Coaches already pay for tools (booking apps, video review); the pitch is "automatic trick-level progress tracking for your whole roster."

Subscription or per-service?

Subscription (per studio, with seats). Pricing suggestion:

Plan	Price	What it includes
Coach Solo	€49 /mo	1 coach, 20 active athletes, full analytics
Gym	€149 /mo	3 coaches, 100 athletes, multi-team cohorts
Gym+	€299 /mo	Unlimited coaches/athletes, white-label reports, SLA

Implementation

Surface: `pole_fe` coach dashboard + multi-tenant security layer. No new ML.

Already built (reuse):

- Cohort z-score + histogram engine (exactly the cross-athlete analytics needed).
- Model registry / training studio (coach can retrain class usage over time).
- Keycloak — org/group support, magic links, temp access, GDPR purge.
- `pole_api` slices (video, training, tools) + jobs infra for async processing.
- Angular frontends with design-system tokens (dashboards come cheap).

Must add (gaps):

- Organization/tenant model in Keycloak (realm groups or org clients).
- Coach dashboard screens: roster management, per-athlete history, alerts.
- Billing per studio (Stripe subscriptions + seat math).
- Role-based gating (coach vs athlete see different scopes).

Effort & priority

Effort: Medium. Multi-tenant scoping + dashboard UX are the bulk; no model work.

Priority: ★★★★★ — highest ARPU per customer of the B2C options and a strong moat (coaches lock in their roster's history).

See also: `docs/app/pole_api/slices.md` , `docs/app/pole_fe/PLAN.md` ,
`docs/app/keycloak/PLAN.md` .

[S3] Deep-Report Upgrade (Freemium)

Type: `subscription/` · freemium → subscription · Priority ★★

Description

The idea in plain English.

Everyone can identify their trick for free — the fun part. Paying users unlock the **understanding** part: a full "deep report" per trick that explains how their execution compares to other athletes, shows the exact frames to fix, and delivers a 4-week training plan written for them.

How it makes money.

Freemium subscription: the free tier feeds the funnel, the paid tier converts:

- **Free** — trick identification + confidence, 3 analyses/week.
- **Deep Report (€7/month)** — unlimited analyses + full deep report.
- **Coach Pack (€14/month)** — deep report + monthly plans + progress history.

Target public.

Athletes who don't just want to know the name of their move — they want to know *why* and *how to improve*. The natural step between "curious beginner" and "serious trainee" who will later pay for the full personal-trainer tier.

What it is

Free users get **trick identification** (the classifier output); paying users get the **deep biomechanical report**: 8-signal histograms, cohort percentile, critical-frame JPEGs, and a 4-week LLM-written plan for a single trick.

How it makes money

Pure freemium upgrade path. The free tier is deliberately *just* identification (exciting, shareable), the paid tier is the *understanding* (repeat value). Converts one trick into a habit; the habit converts into a subscription.

Subscription or per-service?

Subscription (though the same report can also be sold as a per-service item after — see P2 for the one-off version).

Tier	Price	What it includes
Free	€0	Trick ID + confidence, 3 analyses/week
Deep Report	€7 /mo	Unlimited ID + full histogram/cohort/critical-frame report
Coach Pack	€14 /mo	Deep report + 4-week plans + historical trends

Implementation

Surface: frontend paywall (`pole_fe` / `pole_analyst`) + `pole_api` report endpoint. No new ML.

Already built (reuse):

- Histogram/cohort z-score analysis, critical-frame extraction (the report guts).
- `pole_chatbot` coaching prompt (LLM-CF-03) for tips + 4-week plans.
- Keycloak identity + rate limiter (free-tier cap).
- `pole_fe` UI (report chart components already exist).

Must add (gaps):

- Billing (Stripe/Paddle) + tier attribute on the Keycloak session/token (or API JWT claim).
- Paywall guard on the report endpoint (`pole_api` `billing` slice).
- Usage counting per account (free weekly cap, unlimited+ for paid).

Effort & priority

Effort: Low — smallest-scope monetization: gate an endpoint, join the dots in the UI.

Priority: ★★ — a quick win to validate willingness-to-pay before S1/S2.

See also: `docs/packages/pole_ml/PLAN.md` , `docs/app/pole_analyst/PLAN.md` .

[A1] Chatbot Pay-Per-Query (API)

Type: `api/` · metered API, usage-based · Priority ★★★★★

Description

The idea in plain English.

Companies don't need to build AI to offer pole-video analysis in their own products. They plug into our assistant, send a video or a question, and get a conversational analysis back ("that's a shoulder mount — here's what to fix"). They pay **per question answered**, so they only pay for what they use.

How it makes money.

Usage-based billing (per query), starting small and scaling with volume:

- Simple text query: €0.001–0.003
- Query + video analysis: €0.01–0.05
- Full chat session (10 messages): €0.02–0.10

No minimums on the starter plan; volume customers get cheaper rates.

Target public.

Other fitness/sports apps, media platforms, and training companies that want "video analysis" as a feature in their product without hiring an AI team — your idea, sold to developers instead of end users.

What it is

Expose the **already-built pole_chatbot agent** (ReAct + LangGraph variants, guardrails, session management) as a public API. Third parties send a video / query and get back the conversational analysis. You bill **per query**.

How it makes money

Metered usage billing — customers pay for what they consume, no seats, no commitments. This is your lowest-barrier option: the full chatbot backend already exists behind `pole_api`.

Subscription or per-service?

Per-service (metered per query), with prepaid credits or postpaid metering.

Meter	Priced unit	Suggested price
Simple query (text)	1 query	€0.001–0.003
Query + video analysis	1 query	€0.01–0.05
Chat session (10 turns)	1 session	€0.02–0.10

Postpaid: meter monthly, invoice via Stripe Billing usage records.

Prepaid: credit packs with expiry (see P1).

Implementation

Surface: API-only — add an API-key + metering layer around the existing `pole_chatbot` / `pole_api` slices. **No new app, no new ML.**

Already built (reuse):

- `pole_chatbot` — ReAct + LangGraph agents, `GuardrailEngine` (deterministic safety), session schema, tool registry (crop/shift/histogram/similarity), rate limiting, metrics.
- `pole_api` — FastAPI backend (feature-sliced), async job pattern (202 + job_id), WebSocket relays.
- Keycloak — API clients, temp access, scoped tokens.
- Jobs infra (Mongo authority + Redis signal) for heavy queries.
- OmniRoute / OpenAI-compatible router to control LLM token cost per query.

Must add (gaps):

- API-key provisioning + scopes (Keycloak client credentials or service tokens).
- Metering service: count queries/tokens per key → usage DB.
- Stripe Billing (usage records) or Stripe Invoices for prepaid packs.
- Quota enforcement (per-key limits on top of existing rate limiter).
- Usage dashboard for API consumers (read-only portal).

Effort & priority

Effort: Low — this is "put a meter on an endpoint that already exists."

Priority: ★★★★★ — fastest revenue, validates B2B demand, and shares its billing rail with A2 and P1.

See also: `docs/packages/chatbot/PLAN.md`, `packages/chatbot/src/pole_chatbot/guardrails.py`, `docs/app/pole_api/slices.md`.

[A2] B2B Video-Analysis API

Type: `api/` · metered API, usage-based · Priority ★★

Description

The idea in plain English.

Any app can send us a video and receive back, automatically: which trick was performed, how confident the system is, and the proof clips. Our customers integrate this "trick recognition" into their own product with a few lines of code — no model training, no AI experts on their payroll.

How it makes money.

Per-video fees with volume discounts:

- Pay-as-you-go: €0.15–0.20 per video.
- Larger volumes (1k+ videos/month): €0.08–0.12 per video.
- Enterprise (50k+/month): negotiated contract.

Target public.

Sports/fitness apps and platforms whose product would improve with automatic trick recognition or movement analysis, but who can't build the ML themselves. Higher-value bigger contracts than consumer pricing.

What it is

Expose **video analysis as a service**: a third-party app uploads a video and receives identified tricks + confidence + crops + similarity matches. Fitness apps, media, and training platforms integrate it without building any ML.

How it makes money

Metered **per video** (or per minute of processed footage). Bigger margin than chat queries because the input is machine-verifiable ("you consumed 12 analyses") and the output is high-value (a recognized trick with proof frames).

Subscription or per-service?

Per-service (metered per video), tiered with volume discounts.

Volume tier	Price per video	Notes
Starter (pay-as-you-go)	€0.15–0.20	+ setup fee €0
Growth (1k+ videos/mo)	€0.08–0.12	volume discount
Enterprise (50k+/mo)	negotiated	SLA, dedicated capacity

Per-minute-of-footage mixed billing is available for long videos (frame processing is the real cost — Meter that, not the file count).

Implementation

Surface: API-only — extend `pole_api` with a public `analysis` slice + API-key auth + job-backed upload. **No new app, no new ML.**

Already built (reuse):

- `pole_api` video upload + async job pattern (202/job_id, Mongo + Redis jobs, WebSocket progress).
- `pole_ml` classifier + embeddings + Chroma nearest-neighbor fallback.
- `pole_crop` / `pole_tools.video_cutter` (FFmpeg primitives) for clip outputs.
- Keycloak API clients + scopes; Trivy/CI/CD already gates deploys.

Must add (gaps):

- Public API-key auth + per-key quotas (same rail as A1).
- Metering per video + per processing minute (seconds of footage).
- Stripe usage billing or prepaid packs.
- SLA/uptime page + simple status endpoint for enterprise tier.
- Egress/data-retention policy (GDPR purge already built in Keycloak flows).

Effort & priority

Effort: Low. Same billing/metring rail as A1; the analysis pipeline is done.

Priority: ★★ — pairs with A1 as the two "money now" API plays.

See also: `docs/app/pole_api/PLAN.md` , `docs/packages/pole_ml/PLAN.md` ,
`docs/packages/pole_tools/PLAN.md` , `docs/packages/pole_crop/PLAN.md` .

[P1] Pay-Per-Video Analysis Credits

Type: per-service/ · prepaid credits (B2C) · Priority ★★

Description

The idea in plain English.

No subscription, no commitment: an athlete uploads a video, pays a small fee to have that one performance analyzed, and walks away. Perfect for the user who just wants to check a single trick today and doesn't want a monthly plan.

How it makes money.

Prepaid credit packs (1 analysis $\approx$ 1 credit, 2 credits for a deep report):

- **Starter (€9)** — 5 analyses.
- **Popular (€25)** — 15 analyses.
- **Pro (€45)** — 30 analyses.

It front-loads cash (you pay before consuming) and catches the occasional user that subscriptions scare away.

Target public.

Casual or occasional athletes — beginners, hobbyists, people preparing a single routine — who won't subscribe monthly but are happy to pay a few euros when they train. Also a natural entry door for newcomers before they upgrade to a subscription.

What it is

Users upload a video without any subscription and pay per analysis with **prepaid credit packs** — the "I just want to check this one trick" product. The same credits can also buy chatbot deep analyses, sharing the billing rail with A1/A2.

How it makes money

Credits = a durable currency that monetizes infrequent B2C users who won't subscribe. It also front-loads cash (prepaid balances) and reduces billing overhead vs per-invoice metering.

Subscription or per-service?

Per-service (credit packs, no subscription required).

Pack	Price	Credits (1 analysis $\approx$ 1 credit)
Starter	€9	5 analyses (€1.80 each)
Popular	€25	15 analyses (€1.67 each)
Pro	€45	30 analyses (€1.50 each)

Deep-report analyses cost 2 credits; chatbot sessions cost 1–3 depending on length. Expiry (e.g., 12 months) is standard.

Implementation

Surface: `pole_fe` checkout + `pole_api` credit ledger. No new ML.

Already built (reuse):

- Video upload + analysis pipeline (`pole_api` video slice, `pole_ml` classifier, `pole_crop` output, Chroma fallback).
- Keycloak identity (accounts, temp access, GDPR purge).
- Rate limiter in `pole_chatbot` (per-credit spend guard).
- Jobs infra for async analysis tracking.

Must add (gaps):

- Credit ledger (Mongo collection: balance, transactions, expiry).
- Checkout (Stripe Payment Links / Paddle) → credit top-up webhook.
- Spend decrement on each analysis with race-safe reservations.
- UI: credit balance + buy-now modal in `pole_fe`.

Effort & priority

Effort: Low — a ledger + checkout around existing analysis.

Priority: ★★ — good bridge revenue that feeds users toward S1/S3.

See also: `docs/app/pole_fe/PLAN.md`, `docs/app/pole_api/slices.md`, `docs/packages/pole_ml/PLAN.md`.

[P2] Grade / Level Assessment

Type: per-service/ · one-off service (B2C) · Priority ★★

Description

The idea in plain English.

"What level am I?" — an exam you can take at home. The athlete performs the official syllabus moves for a pole grading system, the app evaluates each one automatically, and the result is a report saying their assessed level, with video proof of every move and a clear list of what to train to reach the next level.

How it makes money.

One-off fee per assessment, no subscription:

- Single level assessment: €19–29.
- Full syllabus assessment (every level): €39–59.
- Re-assessment within 3 months: €15–25 (keeps people coming back).

Target public.

Students who want to know their level, prepare for official exams, or prove progress to themselves — plus coaches who use the assessment to place new students into the right class.

What it is

A paid **"What level am I?"** certification: the athlete performs the syllabus moves for a federation grading system (e.g., IPSF/IUPPA), the app detects and rates each trick, and the result is a report stating the assessed level with per-move evidence (critical frames), plus a gap list of what to train next.

How it makes money

One-off fee per assessment (higher than a normal analysis because it is certification-like and includes a curated step-by-step protocol). Repeat buyers re-assess after a few months to see progress — a natural re-purchase loop.

Subscription or per-service?

Per-service, one-off. No subscription required.

Item	Price
Single level assessment	€19–29
Full syllabus assessment (all levels)	€39–59
Progress re-assessment (same athlete, ≤3 mo later)	€15–25

Implementation

Surface: `pole_api` assessment slice + `pole_fe` guided capture flow.

Already built (reuse):

- Classifier + confidence history + debounce (`pole_tools.video_cutter`) — detect each attempted move cleanly.
- Histogram / cohort z-scores + critical-frame extraction — per-move evidence.
- `pole_chatbot` LLM tips — gaps + next steps generation.
- Keycloak identity + temp access (multi-athlete-same-account/studio flow).
- Jobs infra for longer assessment runs.

Must add (gaps):

- **Syllabus/curriculum DB:** move → level → required landmarks, imported per federation. (The biggest add — domain mapping, not modeling.)
- Assessment orchestration: guided capture (record N moves), scoring rules (which levels passed), report assembly.
- Optional badge/PDF certificate output.

Effort & priority

Effort: Medium — new domain logic (syllabus + scoring), but zero new ML.

Priority: ★★ — best defensibility of the per-service group (domain-grounded) and the cleanest story for "pay for a concrete outcome."

See also: `docs/packages/pole_ml/PLAN.md` , `docs/packages/pole_tools/PLAN.md` , `docs/app/pole_fe/PLAN.md` .

[L1] White-Label Studio / Gym License

Type: `licensing/` · B2B license fee · Priority ★★

Description

The idea in plain English.

Studios, gyms, and federations buy the whole platform and run it **under their own name and branding** — their own login page, their own athletes, full recognition + coaching features. They get a professional product overnight with "Powered by us" invisible to their customers.

How it makes money.

Annual B2B license fee:

- **Studio (€1,500/year)** — 1 venue, 2 coaches, up to 100 athletes.
- **Gym Chain (€5,000/year)** — multiple venues, custom branding, SLA.
- **Federation (negotiated)** — full white-label bundle.

Target public.

Training studios, gym chains, and pole federations that want to offer their members a branded digital coaching product but have no AI/software team. High check size, few customers needed to matter.

What it is

License the **entire platform** (recognition, coaching, dashboard) to a studio or coaching brand that runs it under **their own name and branding**. They get their own realm/branded login, their own athletes, and the full analysis stack — you get a recurring license fee and zero user-support burden.

How it makes money

Annual/per-seat license fee to studios, gyms, or federations that want the tool as "their" product without building ML.

Subscription or per-service?

License (B2B), recurring annual. The contract is a license, not metered usage; support/SLA tiers add revenue.

Tier	Annual fee	Includes
Studio	€1,500 /yr	1 realm, 2 coaches, 100 athletes, email support
Gym Chain	€5,000 /yr	Multi-location realms, custom branding, SLA
Federation	negotiated	White-label appstore-ready bundle

Implementation

Surface: infra-heavy — multi-tenant provisioning, branding, and onboarding tooling. No model changes (shared model, per-tenant data isolation).

Already built (reuse):

- Keycloak multi-realm setup, theme/branding pipeline (`phase-6` Stitch restyle) — white-label login is the easy, solved part.
- Design-system tokens → per-tenant theming in Angular.
- Model registry / training studio (per-tenant promoted classes).
- Jobs infra, logging (Filebeat→ES), CI/CD, k3s/Helm for isolated namespaces.

Must add (gaps):

- Tenant provisioning automation (realm + namespace + storage per client).
- Data isolation guarantees + tenant-aware quotas (Mongo/Chroma partitioning).
- Admin/onboarding UI for new tenants (branding upload, athlete import).
- Legal: licensing terms, data ownership/processing agreement (GDPR).

Effort & priority

Effort: High — tenant isolation and onboarding automation dominate the work.

Priority: ★★ — high margin but only makes sense once you have 2-3 committed deal prospects; it is the "enterprise" version of S2.

See also: `docs/app/keycloak/PLAN.md` , `docs/app/infra/PLAN.md` , `docs/app/pole_fe/PLAN.md` .

[L2] Named-Trick Dataset / Kernel Library License

Type: `licensing/` · `data/licensing (B2B)` · Priority ★

Description

The idea in plain English.

We've spent years collecting and processing pole movement data. We sell that data — the analyzed skeletons, movement features, and the digital "fingerprint" of each trick — as a licensed package, so other companies and researchers can build their own movement-analysis products without spending years collecting data.

How it makes money.

One-off license with a yearly refresh option:

- Landmarks + features (50 tricks): €500–1,500.
- Ready-to-use trick library: €1,500–3,000.
- Full bundle + yearly updates: €3,000–5,000/year.

Target public.

Sports-tech companies, app developers, and AI researchers who need real movement data to train their own models. (Requires clean data rights on every video before selling — see the caution in the full doc.)

What it is

Sell a **licensed dataset**: normalized landmark sequences, the per-frame biomechanical features (angles/speeds), and trick embeddings (LSTM bottleneck + Chroma) as a packaged, documented artifact that other apps/researchers buy to bootstrap their own analysis without collecting their own data.

How it makes money

One-off license fees with attribution restrictions; add a maintenance tier for re-fresh releases (new tricks, more athletes).

Subscription or per-service?

One-off license (perpetual or 1-yr), with an optional refresh subscription.

Offering	Price
Landmarks + features (raw, 50 tricks)	€500–1,500
Embedding library + Chroma index (trainable)	€1,500–3,000
Full bundle + refresh subscription (annual)	€3,000–5,000 /yr

Implementation

Surface: data engineering — export pipelines + licensing/delivery. No new ML.

Already built (reuse):

- `pole_tools.process_data` — normalized landmarks/windows (the core artifact).
- `pole_tools.process_embeddings` — embedding/Chroma generation and `manifest.json`.
- Mongo storage + cohort/histogram stats as value-add columns.
- `pole_tools.audit_clips` / `samples_info` for QC of the shipped set.

Must add (gaps):

- Export/catalog tool (schema, versioning, checksums, sample splits).
- **Data-rights legal work** — every video/athlete in the set must be licensed for redistribution (the crawler is Instagram-derived; verify provenance first).
- Anonymization (no athlete identity in shipped landmarks) + GDPR compliance.
- Licensing/payment + delivery (signed URL / package registry).

Effort & priority

Effort: Medium-High (the data-legal part, not the code).

Priority: ★ — real money early on, but depends on data-rights being clean; treat as opportunistic rather than a pillar.

See also: `docs/packages/pole_tools/PLAN.md`, `docs/packages/pole_ml/PLAN.md`, `docs/packages/pole_crawler/PLAN.md` (provenance caution).

[M1] Coach Lead-Generation Marketplace

Type: marketplace/ · two-sided marketplace · Priority ★

Description

The idea in plain English.

An athlete analyzes a trick and the app asks: "why not learn it properly?" — showing coaches and studios near them who teach exactly that move. Athletes stay free; coaches pay to appear in the results and receive new students.

How it makes money.

Coaches are the paying side:

- Per lead (verified contact): €20–50.
- Featured in results (per trick): €50/month.
- Studio profile + unlimited leads: €99/month.

Target public.

Pole studios and independent coaches looking for new students — they already pay for ads and think nothing of a cost-per-lead. Athletes remain free users, which keeps the pool of leads growing.

What it is

After a free trick analysis, the app shows "coaches who teach this trick": studios/coaches pay to appear in those results, and athletes message/visit them. The analysis becomes a **discovery funnel** — "you just landed a shoulder mount; here are 3 local coaches who teach it."

How it makes money

Coaches/studios are the paying side: pay-per-lead, featured listings, or a monthly visibility subscription. Athletes stay free (keeps the funnel big).

Subscription or per-service?

Per-lead or featured listing (per-service for coaches), optionally wrapped in a monthly "visibility" subscription.

Coach option	Price
Per lead (verified contact)	€20–50
Featured in results ("teaches: shoulder mount")	€50 /mo per trick
Studio profile + unlimited leads	€99 /mo

Implementation

Surface: new marketplace app (listing, matching, messaging/lead capture) on top of existing analysis. No new ML (embedding similarity can rank "coach teaches trick X").

Already built (reuse):

- Classifier + embeddings — "which tricks" per analysis (the matching key).
- Keycloak identity, temp access; Angular design system for a fast frontend.
- `pole_api` jobs + data layers.

Must add (gaps):

- Coach/studio accounts + onboarding (separate of athlete accounts).
- Listing + search/matching ("teaches" ↔ "just landed").
- Lead-capture mechanism (email relay or chat).
- Billing for coaches (subscription or per-lead) — the only monetization path.

Effort & priority

Effort: High (marketplaces need both sides seeded; cold-start problem).

Priority: ★ — big upside if a local network exists, but the hardest to launch and most dependent on distribution.

See also: `docs/app/pole_fe/PLAN.md` , `docs/packages/pole_ml/PLAN.md` .

[M2] Verified Athlete Profiles & Badges (Strava-for-Pole)

Type: marketplace/ · freemium profiles · Priority ★

Description

The idea in plain English.

A public profile for athletes where every trick is "verified" by the app — proof, not just claims. The athlete shows the world "I really landed this shoulder mount", gets ranked on leaderboards, and discovers athletes whose style matches hers. Think Strava for pole.

How it makes money.

Freemium subscription:

- **Free** — profile with up to 5 verified badges, no leaderboard.
- **Verified Athlete (€4/month)** — unlimited badges, ranked leaderboard, advanced stats.
- **Team/Studio (€19/month)** — verified profiles for a whole roster.

Target public.

Ambitious athletes who want social recognition for their training, and studios who want to showcase their students' progress as marketing. The virality comes from athletes sharing their "verified" badges.

What it is

Athlete profiles with **verified trick badges**: the classifier confirms a move, and a "VERIFIED" badge is attached to the profile (trick, date, critical-frame proof). Creates a public leaderboard / "most similar athletes" discovery layer powered by embedding similarity.

How it makes money

Freemium profiles: free profiles with limited badges/bio; paid tier unlocks **unlimited verified badges**, custom profile, stats dashboard, and appearance on leaderboards. Optional "verified" re-certification per season.

Subscription or per-service?

Freemium subscription (B2C), with the verified badge as the status good that drives word-of-mouth.

Tier	Price	Includes
Free	€0	Profile, 5 badges, no leaderboard rank
Verified Athlete	€4 /mo	Unlimited badges, ranked leaderboard, advanced stats
Team/Studio	€19 /mo	Roster profiles + badges (see S2)

Implementation

Surface: profile + verification flow in `pole_fe` ; reuse similarity engine.

Already built (reuse):

- Classifier + confidence history + debounce — the actual trick verification.
- Embedding similarity (`find_by_similarity`) — "athletes like you".
- Histogram/cohort z-scores — the "advanced stats" paid tier.
- Keycloak identity + temp access; Angular design tokens.

Must add (gaps):

- Public profile page + badge grants (event-sourced from verified analyses).
- Anti-fraud: reject low-confidence/spoofed videos (confidence threshold, single-session rule).
- Leaderboard computation from cohort stats; moderation (report/remove).
- Billing for the paid tier.

Effort & priority

Effort: Medium — social features + anti-fraud review are the real work.

Priority: ★ — great growth loop (verified = shareable), but only after a trusted analysis product exists; complements S1 rather than standing alone.

See also: `docs/app/pole_fe/PLAN.md` , `docs/packages/pole_ml/PLAN.md` ,
`packages/pole-train-model/src/pole_ml/classifiers/chroma_classifier.py` .

[E1] Virtual Trick Challenges / Competitions

Type: events/ · one-off entry fees + sponsors · Priority ★★

Description

The idea in plain English.

Online challenges organized for the pole community: "30-day handspring challenge" — participants film themselves, the app **verifies** each trick so nobody can cheat, and results/leaderboards are produced automatically. Brands and studios sponsor the event to reach a highly engaged niche audience.

How it makes money.

Two streams per event:

- **Participant entry fee:** €5–15 per challenge.
- **Sponsorship:** brands/studios pay €200–2,000 to title the event.
- **Event toolkit for federations:** €99–499 to run their own challenges.

Target public.

The pole community (athletes who love challenges and certificates), federations/organizers who want to run verified competitions online, and sports brands looking for a passionate, targeted audience.

What it is

Organized online challenges ("30-day handspring challenge", "shoulder-mount streak week") where participants submit videos and the app **verifies each trick** with the classifier + confidence history. Results, leaderboards, and certificates are produced automatically. Brands/studios sponsor the event.

How it makes money

Two revenue streams:

1. **Entry fees** — per-participant, per-event (the event operator keeps a cut).
2. **Sponsorship** — brands pay to title the event and reach a niche audience.

Subscription or per-service?

Per-event, per-service — an event is a discrete product with a price and dates. No recurring billing.

Stream	Suggested price
Participant entry	€5–15 per event
Sponsor (brand/studio)	€200–2,000 per event
Organizer "event toolkit" (B2B for federations)	€99–499 per event

Implementation

Surface: event engine (schedule, submissions, verification runs, results) + capture flow in `pole_fe`. Model reuses the verification pipeline unchanged.

Already built (reuse):

- Real-time recognition + confidence history + debounce (`pole_tools.video_cutter`)
— trick verification per submission.
- Jobs infra (Mongo + Redis) — batch verification of all submissions.
- Keycloak identity + temp access; Angular design system for the event pages.
- Model registry — per-event specific classes/promoted models.

Must add (gaps):

- Event CRUD + submission window + payment per event.
- Verification job per submission with threshold config (strict vs relaxed).
- Leaderboard + certificate generation.
- Anti-cheat (one video per session, temporal metadata checks).

Effort & priority

Effort: Medium — event orchestration is new, but it composes existing building blocks (verify, queue, results) rather than new ML.

Priority: ★★ — strong community/PR value and sponsorship upside; best run in partnership with an existing studio/federation to seed participants.

See also: `docs/packages/pole_tools/PLAN.md`, `docs/packages/jobs/PLAN.md`, `docs/app/pole_fe/PLAN.md`.